

**CARIBBEAN EXAMINATIONS COUNCIL  
CARIBBEAN ADVANCED PROFICIENCY EXAMINATION®**

**BUILDING AND MECHANICAL ENGINEERING DRAWING**

**OPTION A – MECHANICAL ENGINEERING DRAWING AND DESIGN**

**UNIT 2 – Paper 02**

*2 hours 30 minutes*

**17 MAY 2016 (p.m.)**

**GENERAL INFORMATION**

- 1. This paper consists of SIX questions. Each question is worth 20 marks.**
- 2. Answer ALL questions.**
- 3. Silent, non-programmable calculators may be used for this examination.**
- 4. For this examination, each candidate should have:**  
**Two sheets of drawing paper (both sides may be used)**  
**Drawing board and T-square**  
**Drawing instruments**  
**Data sheet B.S.4500 (both hole and shaft basis), provided as an insert**

**INSTRUCTIONS TO CANDIDATES**

- 1. All dimensions given are in millimetres unless otherwise stated.**
- 2. Where scales are not stated, the full size should be applied.**
- 3. All geometrical construction lines MUST be visible on all drawings.**
- 4. When first-angle or third-angle is not specified, the choice of projection is left to the candidate's discretion, in which case the type of projection used MUST be clearly stated.**
- 5. The candidate should use his/her own judgement to supply any dimensions or other details not directly shown on the drawings.**
- 6. The number of each question MUST be written next to the solution.**
- 7. Each candidate MUST enter his/her school code and registration number in the appropriate space provided at the bottom of the drawing paper.**

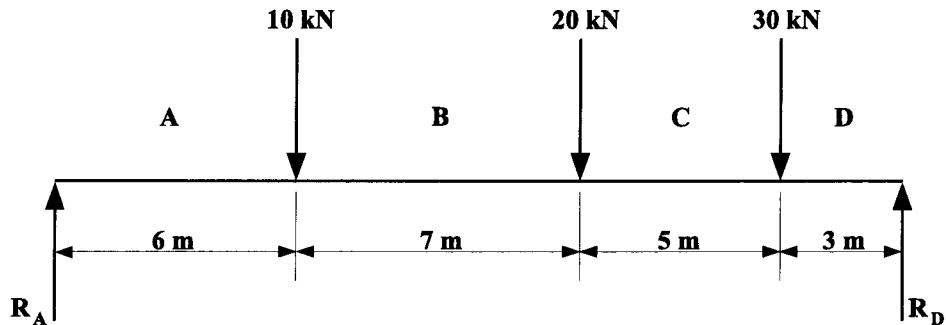
**DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.**

**SECTION A**

**MODULE 1 — MECHANICS OF MACHINES**

**Answer ALL questions from this section.**

1. Figure 1 shows a simple supported beam with three concentrated loads of 10 kN, 20 kN and 30 kN.



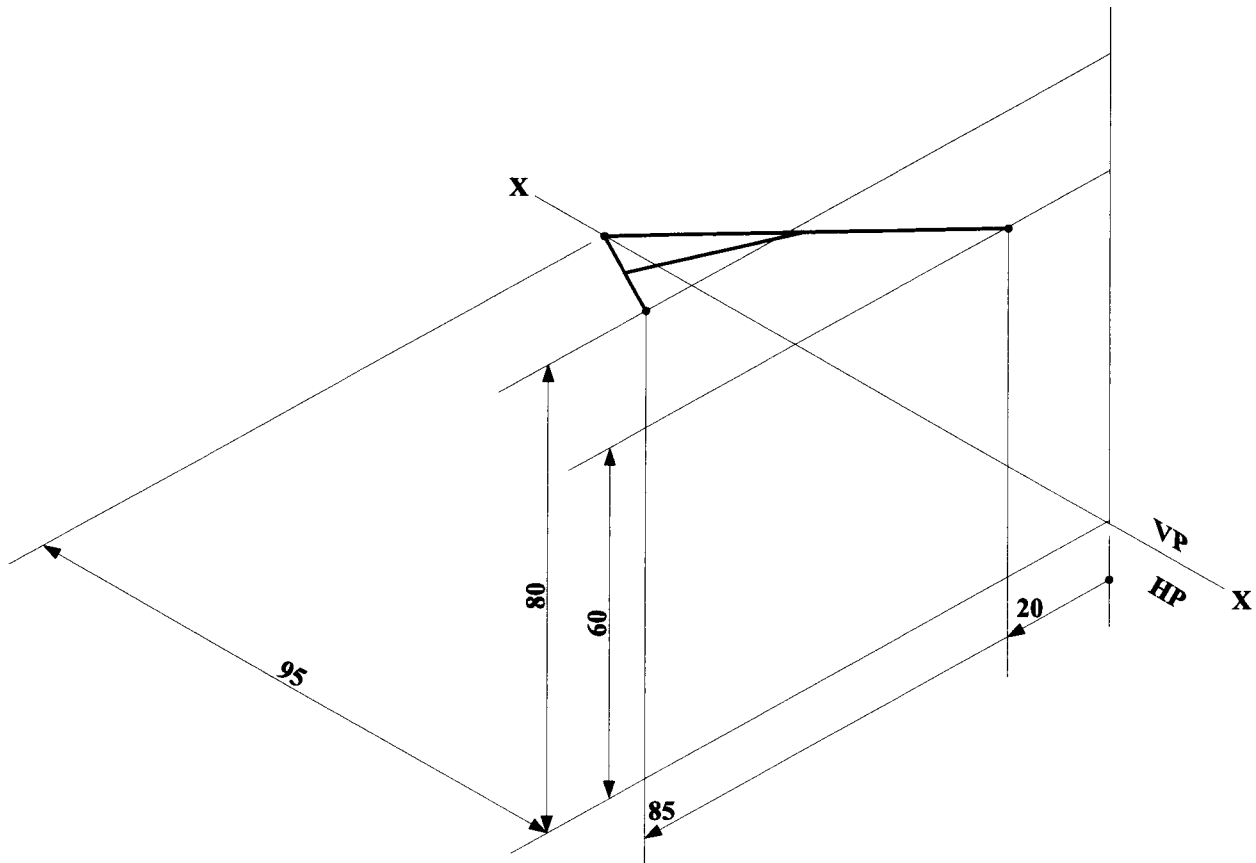
**Figure 1**

- (a) Find, graphically, the magnitude of the reactions  $R_A$  and  $R_D$ . **[15 marks]**
- (b) Verify, analytically, the magnitude of the reactions  $R_A$  and  $R_D$ . **[5 marks]**

**Total 20 marks**

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2. Figure 2 shows a pictorial view of the letter A positioned in space. The cross piece of the letter joins the centre of each side.



**Figure 2**

- |     |                                                               |                   |
|-----|---------------------------------------------------------------|-------------------|
| (a) | Draw the figure in an orthographic view.                      | <b>[7 marks]</b>  |
| (b) | Project auxiliary views of the letter to show its true shape. | <b>[10 marks]</b> |
| (c) | Fully label your drawing using standard notation.             | <b>[3 marks]</b>  |

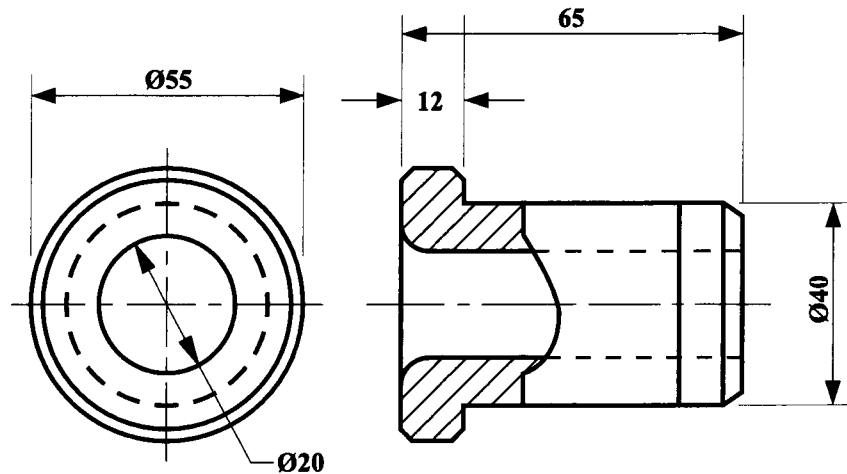
**Total 20 marks**

**SECTION B**

**MODULE 2 — ENGINEERING MATERIALS AND PROCESSES**

**Answer ALL questions from this section.**

3. Figure 3 shows a typical drill bushing with its basic dimensions stated. These bushings are permanently pressed into a drill jig or fixture plate to provide a precise and stable location for drilling. The drill must run freely in the bushing.



**Figure 3**

- (a) Draw the given views of the bushing, selecting suitable dimensions where not provided. **[5 marks]**
- (b) Select appropriate limits and fits for the internal and external dimensions of the bush. The recommended fits are H7/p6 and H8/f7. **[10 marks]**
- (c) Fully dimension the bush, clearly showing ALL the tolerance dimensions. **[5 marks]**

**Total 20 marks**

**GO ON TO THE NEXT PAGE**

4. RAU Design Inc. has undertaken a 'Bearing Project' to design paths for delivering lubrication to bearing surfaces. In this regard, the client has given the company the following four design tasks:
- (i) Single inlet hole in bushing
  - (ii) Circular groove in bushing
  - (iii) Axial groove in bushing
  - (iv) Straight axial groove in the shaft
- (a) Draw sectional views to illustrate any THREE of the above methods for lubricating bearing surfaces. **[6 marks]**
- (b) With the aid of a freehand sketch, illustrate the process of pressurized lubrication by means of a lip seal. **[5 marks]**
- (c) Give TWO side effects of over-lubrication of bearings. **[2 marks]**
- (d) Draw diagrams to illustrate the use of TWO of the following typical ball bearings:
- (i) Radial
  - (ii) Thrust
  - (iii) Angular **[4 marks]**
- (e) State THREE factors that should be considered when designing bearing lubrication. **[3 marks]**

**Total 20 marks**

**GO ON TO THE NEXT PAGE**

## SECTION C

### MODULE 3 — ENGINEERING DESIGN ELEMENTS

Answer ALL questions from this section.

5. Couplings are used to connect machines through rigid shafts. They are flexible to compensate for slight misalignments, vibrations and variations in power transmission.

Figure 4 shows an oldham coupling arrangement which is used to connect two parallel shafts whose axes are a small distance apart. The setup consists of two flanges, each having a rectangular slot, keyed one on each shaft. The two flanges are positioned such that the slot in one is at right angle to the slot in the other.

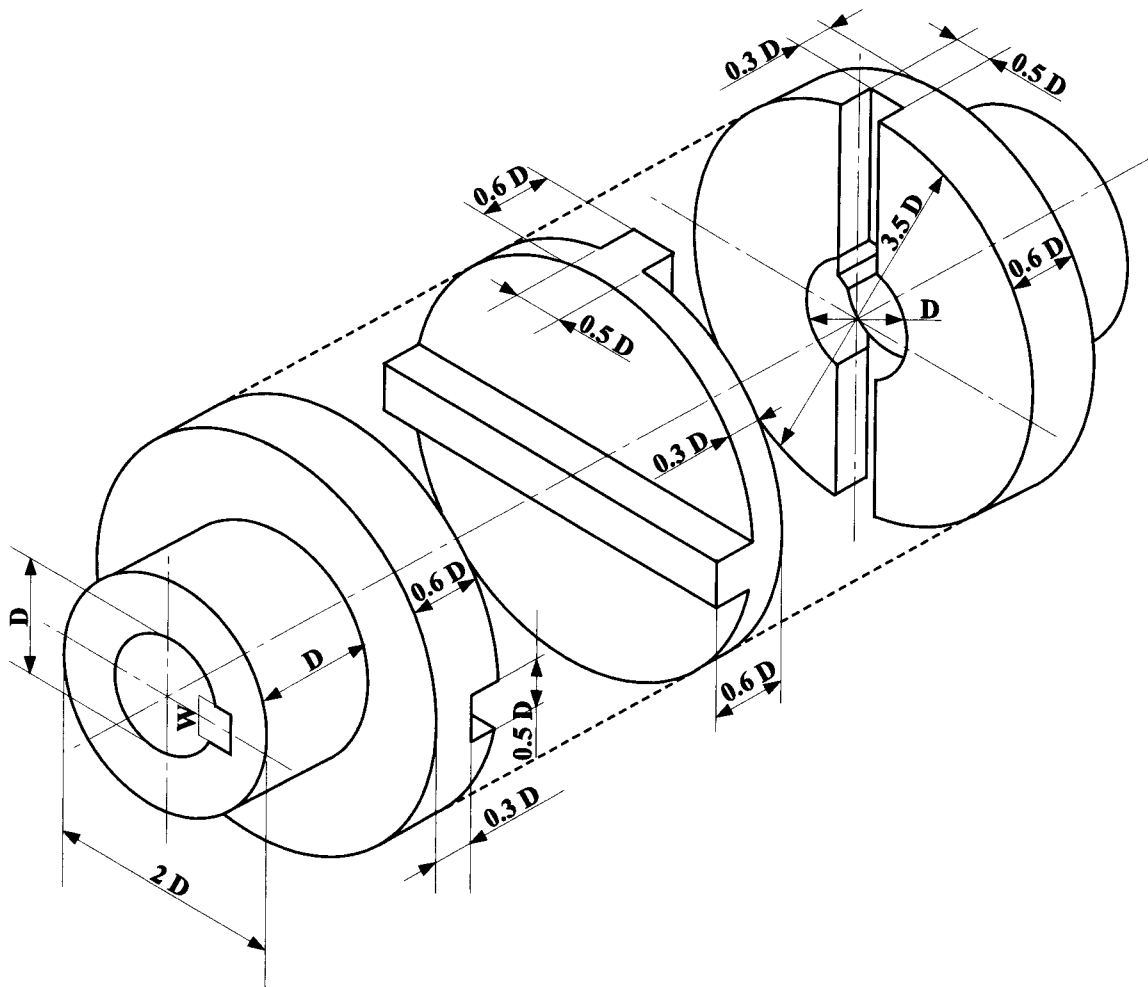


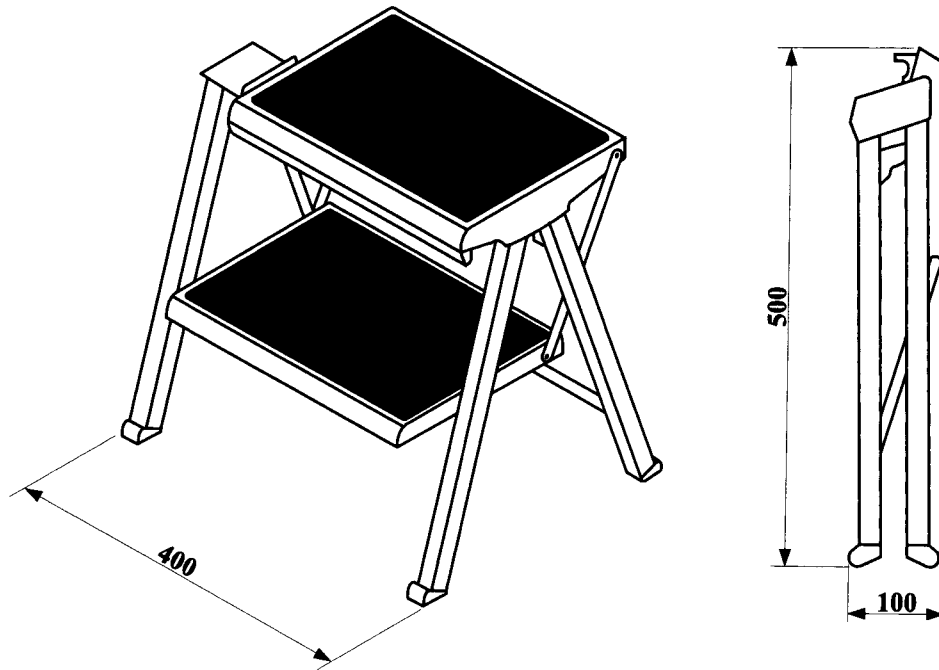
Figure 4

- (a) Draw a front sectional elevation of the assembled coupling. **[6 marks]**
- (b) Draw the side view of the assembled coupling. **[4 marks]**
- (c) Draw a sectional view of a rigid compression coupling and explain its operation. **[9 marks]**
- (d) State ONE safety precaution that should be followed while servicing a clutch assembly. **[1 mark]**

**Total 20 marks**

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6. Figure 5 shows some preliminary ideas for a folding ladder with two steps.



**Figure 5**

- (a) Complete the design of the ladder. Use freehand sketches to illustrate and detail your answer. **[8 marks]**
- (b) Explain your choice of material for use on the ladder. **[2 marks]**
- (c) Explain how the ladder would be manufactured. **[3 marks]**
- (d) Show details of how the steps will be connected to the frame of the ladder. **[2 marks]**
- (e) Explain how the ladder will be able to fold. **[2 marks]**
- (f) List THREE safety precautions that should be recommended for using the ladder. **[3 marks]**

**Total 20 marks**

**END OF TEST**

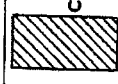
**IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.**



## SELECTED ISO FITS—HOLE BASIS

[illegible]

# BRITISH STANDARD SELECTED ISO FITS—SHAFT BASIS

| Clearance fits                                                                                                                                                                                                                                                                   |     |           |      |           |      |           |      |           |      | Transition fits                                                                                                                                                                              |      |           |     |           |    |           |     |           |     | Interference fits                                                                                                                                                                            |    |           |     |           |     |           |     |           |     |                                                                                                                                                                                            |     |     |  |  |  |  |  |  |  |                                                                                                                                                                                          |  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                                                                                                 |  |  |  |  |  |  |  |  |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----------|------|-----------|------|-----------|------|-----------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------|-----|-----------|----|-----------|-----|-----------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----------|-----|-----------|-----|-----------|-----|-----------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|--|--|--|--|--|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|--|--|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|--|--|--|--|
| <br>+<br><br>D10<br><br>h11 |     |           |      |           |      |           |      |           |      | <br>E9<br>h9<br><br>h9 |      |           |     |           |    |           |     |           |     | <br>F8<br>h7<br><br>h6 |    |           |     |           |     |           |     |           |     | <br>G7<br>h6<br><br>h6 |     |     |  |  |  |  |  |  |  | <br>h6<br>K7<br><br>h6 |  |  |  |  |  |  |  |  |  | <br>h6<br>N7<br><br>h6<br>P7<br><br>h6<br>S7 |  |  |  |  |  |  |  |  |  |
| Nominal sizes                                                                                                                                                                                                                                                                    |     | Tolerance |      | Tolerance |      | Tolerance |      | Tolerance |      | Tolerance                                                                                                                                                                                    |      | Tolerance |     | Tolerance |    | Tolerance |     | Tolerance |     | Tolerance                                                                                                                                                                                    |    | Tolerance |     | Tolerance |     | Tolerance |     | Tolerance |     | Nominal sizes                                                                                                                                                                              |     |     |  |  |  |  |  |  |  |                                                                                                                                                                                          |  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                                                                                                 |  |  |  |  |  |  |  |  |  |
| Over                                                                                                                                                                                                                                                                             | To  | h11       | C11  | h9        | D10  | h9        | E9   | h7        | F8   | h6                                                                                                                                                                                           | G7   | h6        | H7  | h6        | K7 | h6        | N7  | h6        | P7  | h6                                                                                                                                                                                           | S7 | Over      | To  | Over      | To  | Over      | To  | Over      | To  | mm                                                                                                                                                                                         | mm  |     |  |  |  |  |  |  |  |                                                                                                                                                                                          |  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                                                                                                 |  |  |  |  |  |  |  |  |  |
| 3                                                                                                                                                                                                                                                                                | 3   | 0         | +120 | 0         | +60  | 0         | +39  | 0         | +20  | 0                                                                                                                                                                                            | +12  | 0         | +10 | 0         | 0  | 0         | -4  | 0         | -6  | 0                                                                                                                                                                                            | 0  | -14       | 3   | 6         | 3   | 6         | 3   | 6         | 3   | 6                                                                                                                                                                                          | 3   | 6   |  |  |  |  |  |  |  |                                                                                                                                                                                          |  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                                                                                                 |  |  |  |  |  |  |  |  |  |
| 6                                                                                                                                                                                                                                                                                | 6   | 0         | +145 | 0         | +78  | 0         | +50  | 0         | +28  | 0                                                                                                                                                                                            | +16  | 0         | +12 | 0         | 0  | 0         | -4  | 0         | -8  | 0                                                                                                                                                                                            | 0  | -14       | 6   | 10        | 6   | 10        | 6   | 10        | 6   | 10                                                                                                                                                                                         | 6   | 10  |  |  |  |  |  |  |  |                                                                                                                                                                                          |  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                                                                                                 |  |  |  |  |  |  |  |  |  |
| 10                                                                                                                                                                                                                                                                               | 10  | 0         | +170 | 0         | +98  | 0         | +61  | 0         | +35  | 0                                                                                                                                                                                            | +20  | 0         | +15 | 0         | 0  | 0         | -4  | 0         | -9  | 0                                                                                                                                                                                            | 0  | -16       | 10  | 18        | 10  | 18        | 10  | 18        | 10  | 18                                                                                                                                                                                         | 10  | 18  |  |  |  |  |  |  |  |                                                                                                                                                                                          |  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                                                                                                 |  |  |  |  |  |  |  |  |  |
| 18                                                                                                                                                                                                                                                                               | 18  | 0         | +205 | 0         | +120 | 0         | +75  | 0         | +43  | 0                                                                                                                                                                                            | +24  | 0         | +18 | 0         | 0  | 0         | -4  | 0         | -9  | 0                                                                                                                                                                                            | 0  | -19       | 18  | 30        | 18  | 30        | 18  | 30        | 18  | 30                                                                                                                                                                                         | 18  | 30  |  |  |  |  |  |  |  |                                                                                                                                                                                          |  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                                                                                                 |  |  |  |  |  |  |  |  |  |
| 30                                                                                                                                                                                                                                                                               | 30  | 0         | +240 | 0         | +149 | 0         | +92  | 0         | +53  | 0                                                                                                                                                                                            | +28  | 0         | +21 | 0         | 0  | 0         | -5  | 0         | -11 | 0                                                                                                                                                                                            | 0  | -23       | 30  | 40        | 30  | 40        | 30  | 40        | 30  | 40                                                                                                                                                                                         | 30  | 40  |  |  |  |  |  |  |  |                                                                                                                                                                                          |  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                                                                                                 |  |  |  |  |  |  |  |  |  |
| 40                                                                                                                                                                                                                                                                               | 40  | 0         | +280 | 0         | +180 | 0         | +112 | 0         | +64  | 0                                                                                                                                                                                            | +34  | 0         | +25 | 0         | 0  | 0         | -7  | 0         | -13 | 0                                                                                                                                                                                            | 0  | -28       | 40  | 50        | 40  | 50        | 40  | 50        | 40  | 50                                                                                                                                                                                         | 40  | 50  |  |  |  |  |  |  |  |                                                                                                                                                                                          |  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                                                                                                 |  |  |  |  |  |  |  |  |  |
| 50                                                                                                                                                                                                                                                                               | 50  | 0         | +330 | 0         | +220 | 0         | +134 | 0         | +76  | 0                                                                                                                                                                                            | +40  | 0         | +30 | 0         | 0  | 0         | -9  | 0         | -19 | 0                                                                                                                                                                                            | 0  | -39       | 50  | 65        | 50  | 65        | 50  | 65        | 50  | 65                                                                                                                                                                                         | 50  | 65  |  |  |  |  |  |  |  |                                                                                                                                                                                          |  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                                                                                                 |  |  |  |  |  |  |  |  |  |
| 65                                                                                                                                                                                                                                                                               | 65  | 0         | +340 | 0         | +240 | 0         | +140 | 0         | +80  | 0                                                                                                                                                                                            | +44  | 0         | +33 | 0         | 0  | 0         | -9  | 0         | -19 | 0                                                                                                                                                                                            | 0  | -39       | 65  | 80        | 65  | 80        | 65  | 80        | 65  | 80                                                                                                                                                                                         | 65  | 80  |  |  |  |  |  |  |  |                                                                                                                                                                                          |  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                                                                                                 |  |  |  |  |  |  |  |  |  |
| 80                                                                                                                                                                                                                                                                               | 80  | 0         | +390 | 0         | +260 | 0         | +159 | 0         | +90  | 0                                                                                                                                                                                            | +47  | 0         | +35 | 0         | 0  | 0         | -10 | 0         | -22 | 0                                                                                                                                                                                            | 0  | -45       | 80  | 100       | 80  | 100       | 80  | 100       | 80  | 100                                                                                                                                                                                        | 80  | 100 |  |  |  |  |  |  |  |                                                                                                                                                                                          |  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                                                                                                 |  |  |  |  |  |  |  |  |  |
| 100                                                                                                                                                                                                                                                                              | 100 | 0         | +400 | 0         | +280 | 0         | +172 | 0         | +96  | 0                                                                                                                                                                                            | +50  | 0         | +37 | 0         | 0  | 0         | -10 | 0         | -22 | 0                                                                                                                                                                                            | 0  | -45       | 100 | 120       | 100 | 120       | 100 | 120       | 100 | 120                                                                                                                                                                                        | 100 | 120 |  |  |  |  |  |  |  |                                                                                                                                                                                          |  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                                                                                                 |  |  |  |  |  |  |  |  |  |
| 120                                                                                                                                                                                                                                                                              | 120 | 0         | +450 | 0         | +305 | 0         | +185 | 0         | +106 | 0                                                                                                                                                                                            | +54  | 0         | +40 | 0         | 0  | 0         | -12 | 0         | -25 | 0                                                                                                                                                                                            | 0  | -52       | 120 | 140       | 120 | 140       | 120 | 140       | 120 | 140                                                                                                                                                                                        | 120 | 140 |  |  |  |  |  |  |  |                                                                                                                                                                                          |  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                                                                                                 |  |  |  |  |  |  |  |  |  |
| 140                                                                                                                                                                                                                                                                              | 140 | 0         | +460 | 0         | +315 | 0         | +190 | 0         | +112 | 0                                                                                                                                                                                            | +58  | 0         | +42 | 0         | 0  | 0         | -12 | 0         | -25 | 0                                                                                                                                                                                            | 0  | -52       | 140 | 160       | 140 | 160       | 140 | 160       | 140 | 160                                                                                                                                                                                        | 140 | 160 |  |  |  |  |  |  |  |                                                                                                                                                                                          |  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                                                                                                 |  |  |  |  |  |  |  |  |  |
| 160                                                                                                                                                                                                                                                                              | 160 | 0         | +480 | 0         | +330 | 0         | +200 | 0         | +120 | 0                                                                                                                                                                                            | +62  | 0         | +44 | 0         | 0  | 0         | -13 | 0         | -32 | 0                                                                                                                                                                                            | 0  | -56       | 160 | 180       | 160 | 180       | 160 | 180       | 160 | 180                                                                                                                                                                                        | 160 | 180 |  |  |  |  |  |  |  |                                                                                                                                                                                          |  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                                                                                                 |  |  |  |  |  |  |  |  |  |
| 180                                                                                                                                                                                                                                                                              | 180 | 0         | +530 | 0         | +355 | 0         | +215 | 0         | +132 | 0                                                                                                                                                                                            | +66  | 0         | +46 | 0         | 0  | 0         | -14 | 0         | -36 | 0                                                                                                                                                                                            | 0  | -60       | 180 | 200       | 180 | 200       | 180 | 200       | 180 | 200                                                                                                                                                                                        | 180 | 200 |  |  |  |  |  |  |  |                                                                                                                                                                                          |  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                                                                                                 |  |  |  |  |  |  |  |  |  |
| 200                                                                                                                                                                                                                                                                              | 200 | 0         | +570 | 0         | +380 | 0         | +225 | 0         | +144 | 0                                                                                                                                                                                            | +70  | 0         | +48 | 0         | 0  | 0         | -14 | 0         | -36 | 0                                                                                                                                                                                            | 0  | -60       | 200 | 225       | 200 | 225       | 200 | 225       | 200 | 225                                                                                                                                                                                        | 200 | 225 |  |  |  |  |  |  |  |                                                                                                                                                                                          |  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                                                                                                 |  |  |  |  |  |  |  |  |  |
| 225                                                                                                                                                                                                                                                                              | 225 | 0         | +570 | 0         | +400 | 0         | +240 | 0         | +156 | 0                                                                                                                                                                                            | +74  | 0         | +50 | 0         | 0  | 0         | -16 | 0         | -40 | 0                                                                                                                                                                                            | 0  | -68       | 225 | 250       | 225 | 250       | 225 | 250       | 225 | 250                                                                                                                                                                                        | 225 | 250 |  |  |  |  |  |  |  |                                                                                                                                                                                          |  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                                                                                                 |  |  |  |  |  |  |  |  |  |
| 250                                                                                                                                                                                                                                                                              | 250 | 0         | +620 | 0         | +430 | 0         | +265 | 0         | +172 | 0                                                                                                                                                                                            | +78  | 0         | +52 | 0         | 0  | 0         | -16 | 0         | -40 | 0                                                                                                                                                                                            | 0  | -68       | 250 | 280       | 250 | 280       | 250 | 280       | 250 | 280                                                                                                                                                                                        | 250 | 280 |  |  |  |  |  |  |  |                                                                                                                                                                                          |  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                                                                                                 |  |  |  |  |  |  |  |  |  |
| 280                                                                                                                                                                                                                                                                              | 280 | 0         | +650 | 0         | +460 | 0         | +280 | 0         | +184 | 0                                                                                                                                                                                            | +82  | 0         | +54 | 0         | 0  | 0         | -16 | 0         | -40 | 0                                                                                                                                                                                            | 0  | -68       | 280 | 315       | 280 | 315       | 280 | 315       | 280 | 315                                                                                                                                                                                        | 280 | 315 |  |  |  |  |  |  |  |                                                                                                                                                                                          |  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                                                                                                 |  |  |  |  |  |  |  |  |  |
| 315                                                                                                                                                                                                                                                                              | 315 | 0         | +720 | 0         | +500 | 0         | +300 | 0         | +200 | 0                                                                                                                                                                                            | +86  | 0         | +56 | 0         | 0  | 0         | -16 | 0         | -40 | 0                                                                                                                                                                                            | 0  | -68       | 315 | 355       | 315 | 355       | 315 | 355       | 315 | 355                                                                                                                                                                                        | 315 | 355 |  |  |  |  |  |  |  |                                                                                                                                                                                          |  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                                                                                                 |  |  |  |  |  |  |  |  |  |
| 355                                                                                                                                                                                                                                                                              | 355 | 0         | +760 | 0         | +530 | 0         | +320 | 0         | +216 | 0                                                                                                                                                                                            | +90  | 0         | +58 | 0         | 0  | 0         | -17 | 0         | -41 | 0                                                                                                                                                                                            | 0  | -73       | 355 | 400       | 355 | 400       | 355 | 400       | 355 | 400                                                                                                                                                                                        | 355 | 400 |  |  |  |  |  |  |  |                                                                                                                                                                                          |  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                                                                                                 |  |  |  |  |  |  |  |  |  |
| 400                                                                                                                                                                                                                                                                              | 400 | 0         | +840 | 0         | +580 | 0         | +360 | 0         | +240 | 0                                                                                                                                                                                            | +98  | 0         | +60 | 0         | 0  | 0         | -17 | 0         | -41 | 0                                                                                                                                                                                            | 0  | -73       | 400 | 450       | 400 | 450       | 400 | 450       | 400 | 450                                                                                                                                                                                        | 400 | 450 |  |  |  |  |  |  |  |                                                                                                                                                                                          |  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                                                                                                 |  |  |  |  |  |  |  |  |  |
| 450                                                                                                                                                                                                                                                                              | 450 | 0         | +880 | 0         | +600 | 0         | +380 | 0         | +256 | 0                                                                                                                                                                                            | +102 | 0         | +62 | 0         | 0  | 0         | -17 | 0         | -41 | 0                                                                                                                                                                                            | 0  | -73       | 450 | 500       | 450 | 500       | 450 | 500       | 450 | 500                                                                                                                                                                                        | 450 | 500 |  |  |  |  |  |  |  |                                                                                                                                                                                          |  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                                                                                                 |  |  |  |  |  |  |  |  |  |
| 500                                                                                                                                                                                                                                                                              | 500 | 0         | +480 | 0         | +230 | 0         | +135 | 0         | +68  | 0                                                                                                                                                                                            | +20  | 0         | +63 | 0         | 0  | 0         | -17 | 0         | -45 | 0                                                                                                                                                                                            | 0  | -80       | 500 |           | 500 |           | 500 |           | 500 |                                                                                                                                                                                            | 500 |     |  |  |  |  |  |  |  |                                                                                                                                                                                          |  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                                                                                                 |  |  |  |  |  |  |  |  |  |



Holes



Shafts

Diagram to  
scale for  
25 mm diameter